

COURSE BROCHURE 2025-2026



AI & Python Training

Durgapur · Asansol · Online

Build Real AI Skills That Get You Hired

Project-first curriculum · Python → ML → Deep Learning → GenAI → Live Deployment

No prior coding experience required

■ Python Core

■ Data Analysis

■ Machine Learning

■ Deep Learning

■ NLP

■ Gen AI

■ Automation

■ Deployment

300+

STUDENTS TRAINED

9

COURSE MODULES

100%

PROJECT BASED

5.0★

AVG RATING



Python Programming

AI & Python Training — Durgapur

Python is the world's most popular language for AI and data science. We start from absolute basics — no prior coding required. By the end you will write Python programs that process data, automate tasks and solve real problems.

Variables & Data Types

Pandas DataFrames

Lists & Tuples

NumPy Arrays

Dictionaries

File Handling

WHAT YOU WILL LEARN

1

Lists & Tuples

Create, index, slice and manipulate ordered collections. Master list comprehensions — a core Python pattern used everywhere in data science.

2

Strings & Text Processing

Format, split, join and search strings. Build the NLP foundation by understanding how Python handles text — methods, f-strings, regex basics.

3

Dictionaries & Sets

Store key-value data, build lookup tables and process JSON-style records. Dictionaries are the backbone of APIs and real-world datasets.

4

Functions, Loops & File I/O

Write reusable functions, iterate with for/while loops, handle exceptions and read/write CSV and text files — essential for every project.

◆ LEARNING OUTCOMES

- ✓ Write Python programs from scratch with full confidence
- ✓ Process and manipulate real-world data structures
- ✓ Build a student record manager and a CSV report generator
- ✓ Ready to work with Pandas and data science libraries



Before building any AI model, you must understand your data. This module teaches you to load, clean, analyse and visualise real-world datasets using Python's most powerful data science libraries — the same tools used by data scientists at top companies worldwide.

Pandas DataFrames

NumPy Arrays

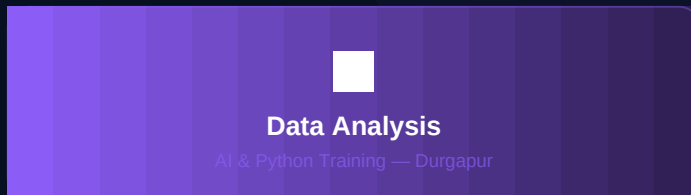
Matplotlib Charts

Seaborn Heatmaps

Data Cleaning

Correlation Analysis

WHAT YOU WILL LEARN



1 Pandas — The Data Workhorse

Load CSV and Excel files into DataFrames. Filter, select, group and aggregate. Handle missing values and merge datasets — bread and butter of every analyst.

2 NumPy — Fast Mathematical Operations

Work with multi-dimensional arrays, perform vectorised calculations at lightning speed. NumPy is the foundation of Scikit-learn, TensorFlow and every ML library.

3 Matplotlib & Seaborn — Visual Storytelling

Create bar charts, scatter plots, histograms, heatmaps and pie charts. Visualise distributions and correlations to find patterns that drive AI model decisions.

◆ LEARNING OUTCOMES

- ✓ Load and clean any real-world CSV dataset
- ✓ Identify patterns and outliers using visual analysis
- ✓ Build a sales performance dashboard and student analytics report
- ✓ Prepare clean datasets ready for machine learning



Machine Learning

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Machine Learning is the engine behind AI. You will train real prediction models on real datasets using Scikit-learn — the industry standard ML library. You will understand how models learn from data and how to evaluate and improve them.

SUPERVISED LEARNING

1 Regression

Predict continuous values — house prices, stock trends, salary estimates. Linear Regression, Ridge, Lasso.

2 Classification

Predict categories — spam or not, disease or healthy. Decision Trees, Random Forest, SVM, KNN.

3 Model Evaluation

Accuracy, precision, recall, F1-score, confusion matrix, cross-validation — know if your model is really working.

UNSUPERVISED LEARNING

4 Clustering

Group similar data without labels. K-Means, DBSCAN — customer segmentation, anomaly detection.

5 Dimensionality Reduction

PCA for visualising and compressing high-dimensional data before feeding into models.

6 Feature Engineering

Transform raw data into powerful features that dramatically improve model accuracy.

♦ LEARNING OUTCOMES

- ✓ Build a live diabetes prediction model deployed on Streamlit
- ✓ Train a stock price forecasting model using real data
- ✓ Understand and tune any Scikit-learn model pipeline end-to-end



Deep Learning

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Deep Learning powers ChatGPT, image recognition, voice assistants and autonomous vehicles. Design, train and evaluate three types of neural networks using TensorFlow and Keras — the tools used by Google, OpenAI and Meta.

[TensorFlow 2.x](#)[Keras API](#)[Backpropagation](#)[Activation Functions](#)[Dropout & BatchNorm](#)[Transfer Learning](#)

THREE NEURAL NETWORK ARCHITECTURES

ANN

Artificial Neural Networks — The Foundation

Understand neurons, layers, weights and biases. Train a multi-layer perceptron on tabular data. Learn forward propagation, loss functions (MSE, Cross-Entropy) and optimisers (Adam, SGD). Build your first neural network from scratch.

CNN

Convolutional Neural Networks — Computer Vision

Teach machines to see. Use convolutional and pooling layers to extract visual features from images. Build an image classifier and an emotion detection model. Apply transfer learning with VGG and MobileNet.

RNN

Recurrent Neural Networks — Sequential Data

Process time-series and sequential data. LSTM and GRU networks for stock price prediction, text generation and time-series forecasting. Understand the vanishing gradient problem and how LSTMs solve it.

✦ LEARNING OUTCOMES

- ✓ Build an emotion detection CNN — deployed on Google Colab
- ✓ Train an RNN for stock price prediction using LSTM networks
- ✓ Apply transfer learning to solve image classification in hours, not days



Natural Language Processing

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MODULE 05 — NATURAL LANGUAGE PROCESSING

Teach machines to read, understand and classify human language using Python NLP libraries.

1 Text Preprocessing

Tokenisation, stopwords removal, stemming, lemmatisation. Clean raw text into structured inputs for ML models.

2 Sentiment Analysis

Classify text as positive, negative or neutral. Train models using TF-IDF and word embeddings on review data.

3 Text Classification & Chatbots

Build intent classifiers, named entity recognition systems and a rule-based chatbot using NLTK and spaCy.

NLTK

spaCy

TF-IDF

Word2Vec

◆ LEARNING OUTCOMES

- ✓ Build a Bengali-English chatbot using NLTK
- ✓ Create an AI-powered Python app using OpenAI/Gemini APIs
- ✓ Build a custom RAG assistant on your own documents

Generative AI Integration

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MODULE 06 — GENERATIVE AI INTEGRATION

Integrate cutting-edge LLMs and Gen AI APIs into your Python projects — the most in-demand skill of 2025.

1 LLM API Integration

Connect to OpenAI GPT, Google Gemini and open-source models via API. Build AI-powered Python apps in minutes.

2 Prompt Engineering

Design effective prompts for reliable AI outputs. System prompts, few-shot examples, chain-of-thought techniques.

3 RAG & Custom AI Assistants

Build Retrieval-Augmented Generation apps — AI assistants that answer questions from your own documents.

OpenAI API

Gemini

LangChain

RAG



AI Automation

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MODULE 07 — AI AUTOMATION

Automate repetitive tasks using Python. Save hours of manual work every week building smart automation tools.

1 WhatsApp & Email Automation

Send automated WhatsApp messages and personalised emails with Python — bulk messaging, scheduled reports.

2 Task Scheduling & Web Scraping

Schedule Python scripts to run automatically. Scrape data from websites using BeautifulSoup and Selenium.

3 API Integration

Connect Python to external APIs — weather, payments, social media, databases, and real-time integrations.

Twilio, SMTP, Schedule, BeautifulSoup

◆ LEARNING OUTCOMES

- ✓ Build a WhatsApp bulk message automation system in Python
- ✓ Create an interactive ML web app with Streamlit in under an hour
- ✓ Connect live ML models to a user interface with real predictions



Streamlit Web Apps

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MODULE 08 — PROJECTS THROUGH STREAMLIT

Streamlit transforms Python scripts into beautiful interactive web apps — no HTML, CSS or JavaScript required.

1 Streamlit Fundamentals

Widgets, layouts, charts, file uploaders and data tables. Build your first ML app in under 30 minutes.

2 Connecting ML Models to UI

Load saved Scikit-learn / TensorFlow models. Create interactive prediction interfaces with live results.

3 Multi-page AI Applications

Build professional multi-page Streamlit apps with navigation, user

Streamlit, Pickle Models, Custom Styling, Session State



Cloud Deployment

AI & Python Training — Durgapur

This is what makes our course unlike any other training in Durgapur and Asansol. You don't just build projects on your laptop — you deploy them live on the internet with a real URL you can share with anyone in the world, including employers.

GITHUB — VERSION CONTROL & PORTFOLIO

1 Git Fundamentals

Init, commit, push, pull, branch and merge. Every professional developer uses Git daily — you will too.

2 GitHub Repository

Create a professional GitHub profile with pinned AI projects. Your GitHub IS your developer portfolio.

3 README & Documentation

Write professional project documentation. A well-documented project on GitHub impresses employers significantly more than undocumented code.

STREAMLIT CLOUD — LIVE DEPLOYMENT

1 requirements.txt & Config

Prepare your app for cloud deployment. Define dependencies, manage secrets and configure environment settings.

2 One-Click Deployment

Connect GitHub repo to Streamlit Cloud. Every git push auto-deploys your latest code — your live app updates instantly.

3 Custom Domain & Sharing

Get a public URL for your AI app. Share it on your resume, LinkedIn and with employers — proof of real-world skills.

■ LIVE PROJECTS YOU WILL DEPLOY

■ Diabetes Predictor

diabetespredictor2026
.streamlit.app

■ Cab Booking App

cabdgp.streamlit.app

■ Stock Predictor

Google Colab Notebook

■ Emotion Detector

Google Colab Notebook

MEET YOUR INSTRUCTOR
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Biswarup Mukherjee

Python & AI Practitioner · Founder, AI & Python Training

Biswarup Mukherjee

■ Python & AI Practitioner · Educator · Live Deployment Expert



Biswarup Mukherjee is a Python and AI practitioner based in Durgapur, West Bengal, with hands-on experience building and deploying real machine learning applications. He founded AI & Python Training to bridge the gap between theoretical education and industry-ready skills in the Durgapur–Asansol belt. His philosophy: every concept must produce working code. Students don't just learn — they build live, deployed projects they can show to employers from day one.

QUALIFICATION

PG Diploma — AI & Machine Learning

INSTITUTION

National Institute of Technology, Warangal

LOCATION

■ Durgapur & Asansol, West Bengal

SPECIALISATION

Python · Pandas & NumPy · Scikit-learn · TensorFlow / Keras · Streamlit · Flask · NLP & NLTK · Gen AI / LLM APIs · Cloud Deployment
Live AI Deployment · TensorFlow · WhatsApp Automation · GitHub

300+

STUDENTS TRAINED

9

COURSE MODULES

4+

LIVE AI APPS DEPLOYED

5.0★

GOOGLE RATING

Simple, Transparent Pricing

No hidden costs. You will pay back for life.

■ MOST POPULAR

STARTER

₹7,500

Levels 1–2 · Python + Data

- ✓ Python Core (Level 1)
- ✓ Data Handling & Viz (L2)
- ✓ 3 Mini Projects
- ✓ Personal Guidance
- ✗ ML / AI Modules
- ✗ Cloud Deployment

Enroll Now

FULL PROGRAM

₹20,000

All 7 Levels · Complete AI Path

- ✓ All 7 Course Levels
- ✓ Python → Deep Learning → NLP
- ✓ Automation & APIs
- ✓ Live Cloud Deployment ■
- ✓ Portfolio Projects
- ✓ WhatsApp Support

Enroll Now

ML & AI TRACK

₹10,000

Levels 3–7 · ML to Deployment

- ✓ Machine Learning (L3)
- ✓ Deep Learning (L4)
- ✓ NLP & Text AI (L5)
- ✓ Automation (L6)
- ✓ Live Deployment (L7) ■
- ✗ Python Fundamentals

Enroll Now

Get in Touch & Enroll Today

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LOCATIONS
Asansol & Durgapur · Online via Zoom

Ready to Start Your AI Journey?

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